Weighted Proj bridge: checked homogeneity and chart equations
Pinned Lean 4.34.1; Mathlib d13f23b723b8a846827a245b89c10fc7d3f11612

Weights: X=1, Y=3, Z=1. Coefficients a3,a2,a1,a0 are arbitrary in a field.
Equation: Y^2 - X^5 Z - a3 X^3 Z^3 - a2 X^2 Z^4
          - a1 X Z^5 - a0 Z^6.

CHECKED: equation_weighted_homogeneous, weighted degree six.
CHECKED: relationIdeal_homogeneous, principal ideal homogeneous.
CHECKED: Z=1 gives y^2 - (x^5+a3*x^3+a2*x^2+a1*x+a0).
CHECKED: X=1 gives w^2-z(1+a3*z^2+a2*z^3+a1*z^4+a0*z^5).
Compiler exit: 0. Four axiom audits: propext, Classical.choice, Quot.sound.
No sorryAx in any audit.

OPEN: grading on the quotient by the homogeneous ideal.
OPEN: degree-zero localization at Z equals the affine coordinate ring.
OPEN: infinity chart and overlap comparison with constructed two-chart curve.
OPEN: smooth projective completion and Pic0 transfer.

The graph and CSV enumerate these obligations without a ranking claim.
Compiler receipt: weighted-proj-head-final-build.log
Chart receipt: weighted-proj-charts-build.log
Copy timeout receipt: weighted-proj-head-copy.txt